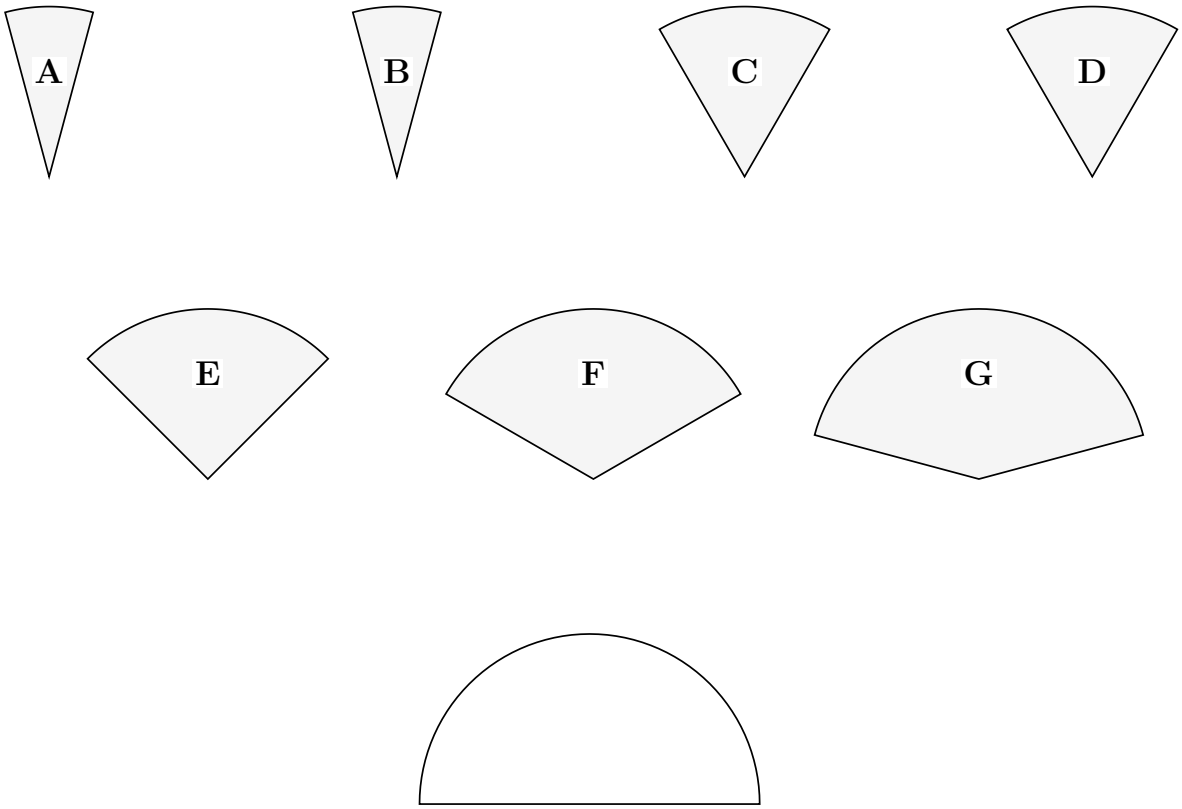
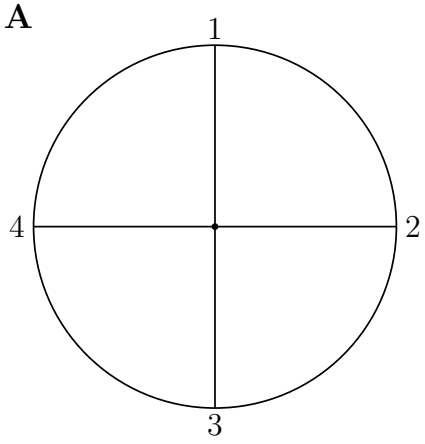
For folds, use the marked front: M means mountain and V means valley. Fold every ray in the finished pattern fully flat. Keep the disk whole, with no other creases.

Problem 1: Use the seven loose wedges to make half-turns. Find as many different groups of letters as you can; use each wedge at most once in a group.

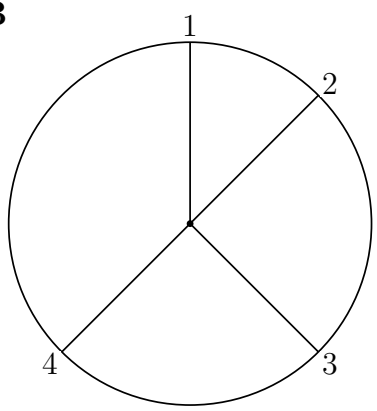


Problem 2: Which of these disks can close flat along every ray? Make one flat fold of each disk that can close and mark its rays M or V.

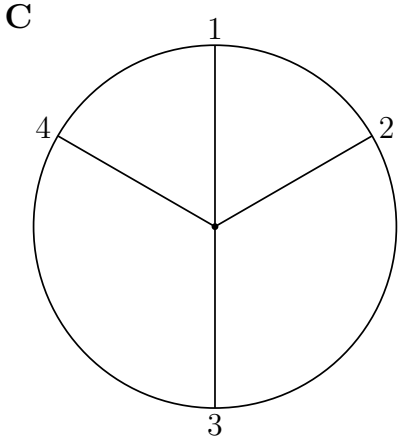
A



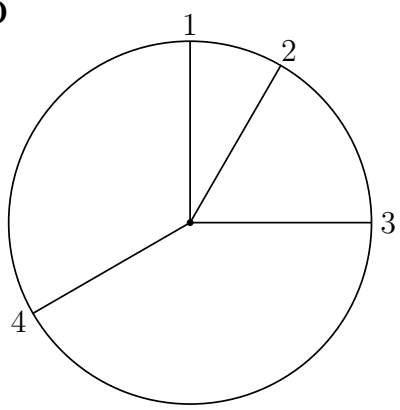
B



C

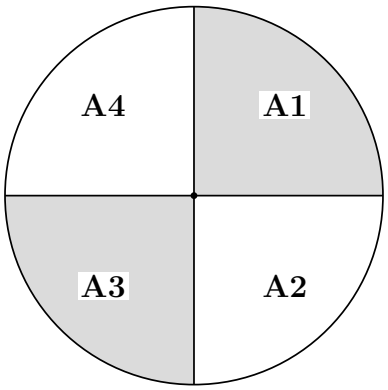


D



Problem 3: Use the 16 separate loose wedges cut to match A1–A4, B1–B4, C1–C4, and D1–D4. For each disk, can the shaded pair fill a half-turn exactly, and can the white pair? For any disk that could not close in Problem 2, explain why.

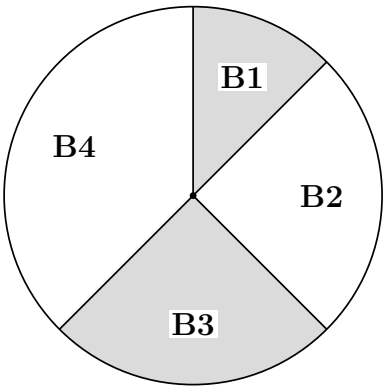
A



Shaded: _____

White: _____

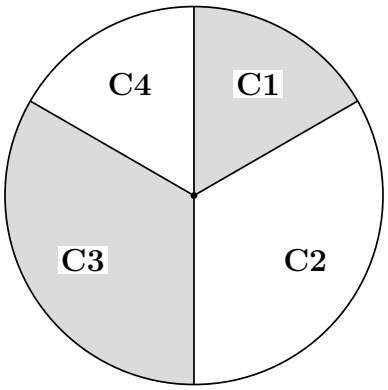
B



Shaded: _____

White: _____

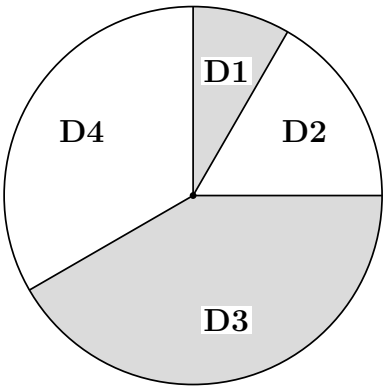
C



Shaded: _____

White: _____

D

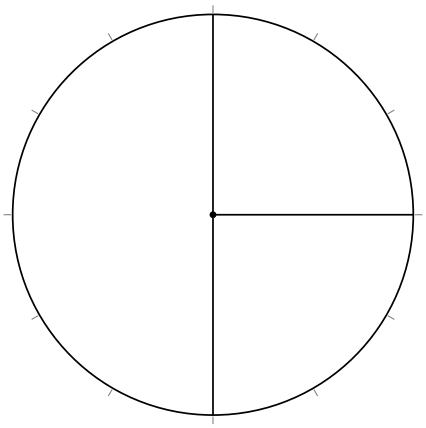


Shaded: _____

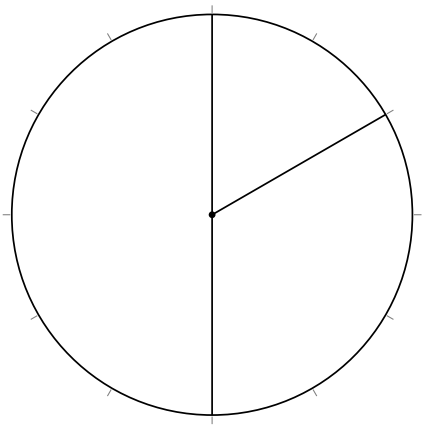
White: _____

Problem 4: Add one ray to each pattern so it can close flat along every ray. Make a flat fold of each completed pattern.

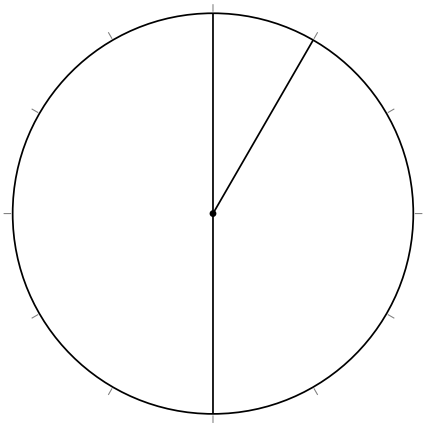
A



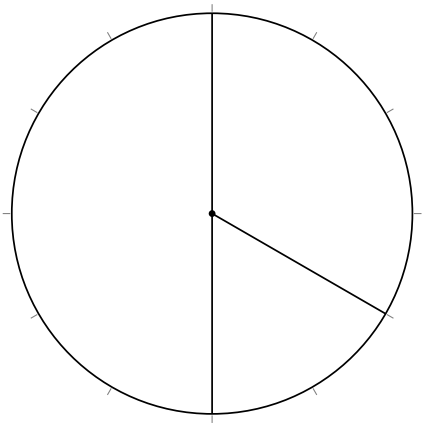
B



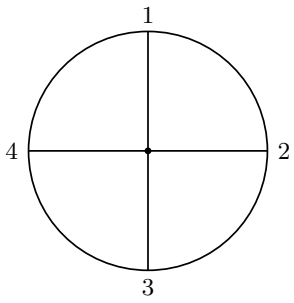
C

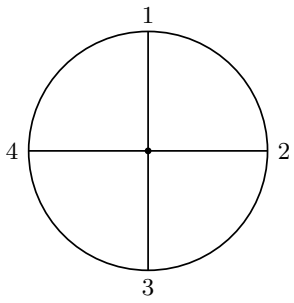


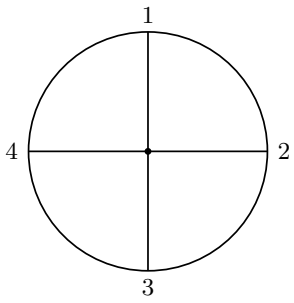
D

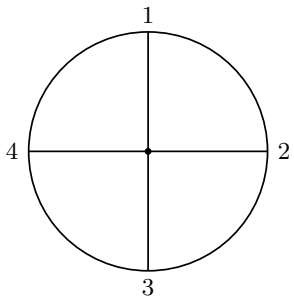


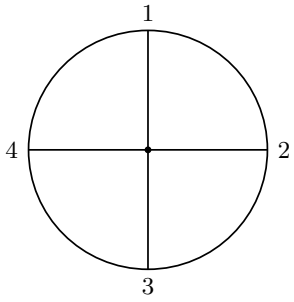
Problem 5: Find every mountain/valley labeling of the four-right-angle disk that can close flat. Keep ray 1 at the top and make each fold on your disk.

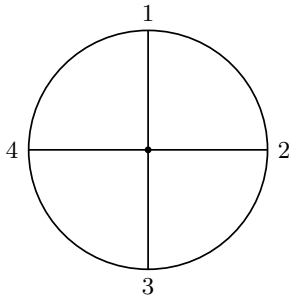


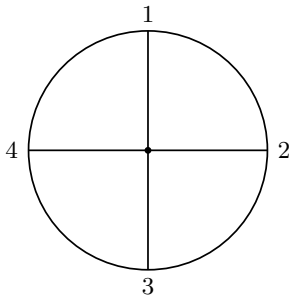


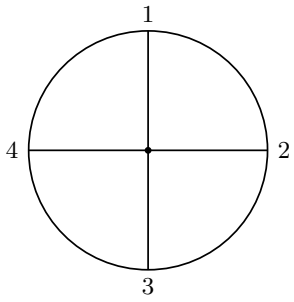


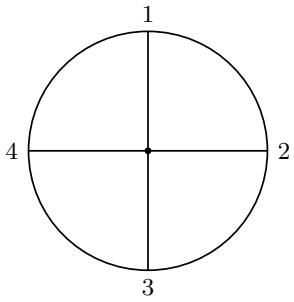


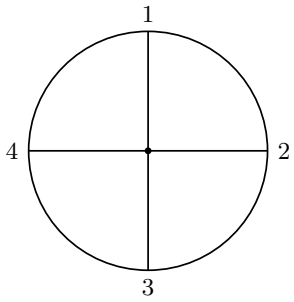












Problem 6: Use these six wedges once each to fill a disk. Find four different orders of A, B, and C in which the alternating wedges make two half-turns. Read clockwise from the top ray.

